

Streaming Joins Revisited: A Probabilistic Perspective

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Abstract

This paper reports a systematic evaluation of sparse graph embeddings. We release the processing scripts and intermediate artefacts to support independent verification. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate.

1. Introduction

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We release the processing scripts and intermediate artefacts to support independent verification. The reduced model retains most of the explanatory power at a fraction of the computational cost. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The measurements reported here were collected over twelve independent trials under controlled conditions.

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. Neither of the alternative formulations improved on the simple additive model in our setting. Longer observation windows narrow the confidence interval but do not shift the central estimate.

Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Taken together, these findings argue for treating the two regimes separately in future analyses.

2. Related Work

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3. Materials and Methods

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4. Experimental Setup

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5. Results

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6. Discussion

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